

Semester Project Specification

Project Title: CryptoSim Pro – Web-Based Cryptocurrency Trading Platform

Duration: 4 Weeks

Team Size: 4 Students

Deadline : 23/07/2026

1. Project Context

A fintech startup wants to launch a cryptocurrency trading platform for new investors. Before investing in a real product, the company requires a fully functional prototype that allows users to trade cryptocurrencies using virtual money.

Your team must design and implement a complete web application that simulates the behavior of a modern trading platform. The system must use live cryptocurrency prices and provide portfolio management, transaction history, performance monitoring, and competitive rankings.

2. Project Objectives

By the end of this project, students should be able to:

- Develop a complete web application.
- Design and implement a database.
- Integrate external REST APIs.
- Build a modern responsive user interface.
- Implement authentication and authorization.
- Apply software engineering best practices.
- Collaborate using Git and GitHub.

3. Mandatory Functional Requirements

FR1 – User Registration and Login

- Register using username, email and password.
- Secure password storage.
- Login and logout functionality.

FR2 – User Dashboard

- Display available cash.
- Display total portfolio value.
- Display gain/loss indicators.
- Display recent transactions.

FR3 – Live Market Data

- Retrieve live cryptocurrency prices from CoinGecko.
- Support at least 10 cryptocurrencies.

FR4 – Trading System

- Buy cryptocurrencies.
- Sell cryptocurrencies.
- Validate available balance.
- Validate owned quantities.
- Apply a 0.5% transaction fee.

FR5 – Portfolio Management

- Display holdings.
- Display asset allocation.
- Calculate unrealized profit/loss.

FR6 – Transaction History

- Store all transactions.
- Filtering and sorting options.

FR7 – Leaderboard

- Rank users by portfolio performance.
- Display Top 10 users.

FR8 – Watchlist

- Users can save favorite cryptocurrencies.

FR9 – Price Alerts

- Users can create custom price alerts.

4. Required Website Pages

1. Landing Page
2. Login Page
3. Registration Page
4. Dashboard
5. Market Page
6. Trading Page
7. Portfolio Page
8. Transaction History Page
9. Leaderboard Page
10. User Profile Page

All pages must share a consistent design and navigation menu.

5. User Interface Requirements

- Responsive design.
- Professional appearance.
- Navigation bar.
- Form validation.
- Loading indicators.
- Error messages.
- Data tables.
- Charts
- Mobile-friendly layout.

6. Technical stack

Backend:

- Python Flask

Frontend:

- HTML5
- CSS3
- JavaScript

Database:

- SQLite or PostgreSQL

External API:

- CoinGecko

Version Control:

- GitHub repository required

8. Team Organization

Each team member must have a clearly defined role.

All members must contribute to integration and documentation.

9. Project Milestones

Week 1

- Requirements analysis
- Wireframes
- Database design
- Project setup

Week 2

- Authentication
- Database implementation

- API integration

Week 3

- Trading system
- Portfolio management
- Leaderboard

Week 4

- Testing
- Bug fixes
- Documentation
- Final presentation

10. Deliverables

1. Source Code

- Complete GitHub repository

2. Technical Report (10–15 pages)

Must include:

- Introduction
- System architecture
- Database design
- Screenshots
- Testing strategy
- Challenges encountered
- Conclusion

3. Presentation video

- 5min max

11. Evaluation Rubric (100 Points)

Functionality: 35

Database Design: 15

UI/UX Quality: 15

Code Quality: 10

API Integration: 5

Testing: 5

Documentation: 10

Presentation: 5

12. Bonus Features (+10 Points Maximum)

Choose one:

- News Feed

- Email Notifications
- AI-Based Trading Recommendations

13. Academic Integrity

All code must be written by the team. External libraries are permitted but must be properly referenced.